NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
MID - SEM EXAMINATION, 2024
SESSION: 2023 – 2024 (Spring)
M.Tech. 2nd Semester and Ph.D

Subject code: CS6218	**Subject Name: Machine Learning**	Dept. Code: **CS**
No. of pages: 2	Full Marks: 30	Duration: **2 Hrs**

Answer ALL questions and All parts of a question should be answered at one place. Unnecessary writings will attract negative marks.

1. a). Find the probability of misclassification of Bayes' decision rule with decision boundary $\Omega_1 = (0.5, 1.5)$ and $\Omega_2 = (1.5, 3)$ for the given example. The class conditional probability density functions of both the classes are as follows: **[4+2]**

$$p_1(x) = x;\ 0 < x < 1$$
$$= 2 - x; 1 \leq x < 2$$
$$= otherwise$$
$$p_2(x) = x - 1;\ 1 < x < 2$$
$$= 3 - x; 2 \leq x < 3$$
$$= otherwise$$

The prior probability of 1st class is P and 2nd class is $(1 - P)$.The decision rule is defined: $\Omega = [0, 3]$

b). In a heads or tails coin-tossing experiment the probability of occurrence of a head (1) is q and that of a tail (0) is $1 - q$. Let $x_i, i = 1, 2, 3, \dots, N$, be the resulting experimental outcomes, $x_i \in \{0, 1\}$. Show that the ML estimate of q is

$$q_{ML} = \sum_{i=1}^{N} x_i$$

The likelihood function is given as follows

$$P(X : q) = \prod_{i=1}^{N} q^{x_i}(1 - q)^{(1-x_i)}$$

2. a). Solve the following classification problem with the perceptron rule. Apply each input vector in order, for as many repetitions as it takes to ensure that the problem is solved. Draw a graph of the problem only after you have found a solution. Where p denotes data points and t denotes target output. **[4+2]**

$$\left\{p_1 = \begin{bmatrix}2\\2\end{bmatrix}, t_1 = 0\right\} \left\{p_2 = \begin{bmatrix}1\\-2\end{bmatrix}, t_2 = 1\right\} \left\{p_3 = \begin{bmatrix}-2\\2\end{bmatrix}, t_3 = 0\right\} \left\{p_4 = \begin{bmatrix}-1\\1\end{bmatrix}, t_4 = 1\right\}$$

Use the initial weights and bias: $W(0) = [0\ 0]$ and $b\ (0) = 0$

b). The backpropagation algorithm is used to compute the gradient of the squared error with respect to the weights and biases of a multilayer network. Write down the expression of error gradient for the neurons in the output layer and error gradient for the neurons in the hidden layer.

3. a). Given the following 2D data points

$\left\{\binom{2}{2}, \binom{2}{-2}, \binom{-2}{-2}, \binom{-2}{2}\right\} \in\ C1;\ \left\{\binom{1}{1}, \binom{1}{-1}, \binom{-1}{-1}, \binom{-1}{1}\right\} \in\ C2$ and the mapping function

$$\emptyset \binom{x_1}{x_2} = \begin{pmatrix} x_1 \\ x_2 \\ \frac{({x_1}^2 + {x_2}^2) - 5}{3} \end{pmatrix}$$

Design a SVM for an optimized hyperplane. **[3+3]**

b). Two training samples are given as follows:

Class A: $x_1 = 1, x_2 = 1\ and\ y = 1$

Class B: $x_1 = -1, x_2 = -1\ and\ y = -1$

$$f(X) = w_1x_1 + w_2x_2 + C \begin{cases} \geq 0\ (Class\ A) \\ < 0\ (Class\ B) \end{cases}$$

Solve w_1, w_2 and C to determine classifier.

4. a). Consider the example below where the mass, $y(Grams)$, of a chemical is related to the time, $x\ (Seconds)$, for which the chemical reaction has been taking place according to the Table 1: **[4+2]**

Table 1: Dataset with Time Vs Mass

Observation Nos	**Time, $x\ (Seconds)$**	**Mass $y(Grams)$**
1	5	40
2	7	120
3	12	180
4	16	210
5	20	240

Apply Least Square Estimation technique to find the equation of the regression line that fits the data best. Illustrate scatter diagram with the regression line. What is the mass of the chemical after ten seconds has passed? How much time does it take for the weight of the chemical to increase by 50 grams?

b). What is Cross Validation method? What is its significance with respect to supervised classification algorithms?

5). a). State the main criterion function with proper explanation for K-NN classifier. In that case consider the following scenario: Consider the one-dimension dataset shown in Table 2. Classify the data point $x = 5.0$ according to its 1-, 3-, 5-, and 9-nearest neighbors (using majority vote). **[3+3]**

Table 2: Dataset Description

x	0.5	3.0	4.5	4.6	4.9	5.2	5.3	5.5	7.0	9.5
y	-	-	+	+	+	-	-	+	-	-

b). Explain the fundamental difference between PCA and LDA with graphical illustration. Compute the Linear Discriminant projection for the following two- dimensional dataset. Find the optimal projection vector.

Samples for Class $w_1: X_1 = (x_1, x_2) = \{(1,2), (2,3), (3,4.9)\}$

Samples for Class $w_2: X_2 = (x_1, x_2) = \{(2,1), (3,2), (4,3.9)\}$

**************************** ALL THE BEST *************************************